README FILE

Password Strength Analyzer

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PASSWORD STRENGTH ANALYZER

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1. PROJECT DESCRIPTION

Password Strength Analyzer is a beginner-level Python cybersecurity project. It checks a test password's length and character variety, gives a basic strength rating, and suggests improvements. The password is processed locally and is not saved or sent over a network.

2. FEATURES

- Checks length (8 characters minimum; recommends 12 or more).
- Checks uppercase and lowercase letters.
- Checks digits and special characters.
- Warns about a small list of commonly guessed passwords.
- Shows a Weak, Moderate, or Strong rating.
- Does not store the entered password.

3. TECHNOLOGIES

- Python 3.8 or newer
- Python standard-library modules: re and getpass
- Optional: Jupyter Notebook

4. HOW TO RUN

Save the Python source as:

Harishankar_AS_PasswordStrengthAnalyzer.py

Run from a terminal:

python Harishankar_AS_PasswordStrengthAnalyzer.py

You can also copy the code into a Jupyter Notebook cell and run it.

5. TESTING

Use only made-up test passwords. Do not enter a real password that protects an account.

6. LIMITATIONS

This is a simple educational checker. Passing its checks does not guarantee a password is secure. It does not query breached-password databases and uses only a small common-password list.

7. SECURITY TIPS

- Use a unique password or passphrase for every account.
- Prefer long passwords/passphrases.
- Use a password manager.
- Enable multi-factor authentication (MFA).
- Never share passwords or reuse a demonstration password on real accounts.